

M O R S

Project Guide

Every file, and what it is for

ASI-HACK Space Analytics Engine - MORS Scientific Command Center

Files 36

Source 15552 lines of Python and HTML

Tree size 1861.9 kB

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How to use this guide

Read it top to bottom once, then keep it open next to the code. Every file gets one card with the same five blocks: role, what it does, inputs and outputs, dependencies, and where in the product you can see its effect.

Python files also list their top-level symbols with line numbers, so the card doubles as an index.

Contents

1. The repository at a glance
2. How a request flows end to end
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6. tools/
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8. data/
9. docs/
10. Regenerating the documents

1. The repository at a glance

36	1861.9 kB	11013	4539
Files	Tree size	Python lines	HTML lines

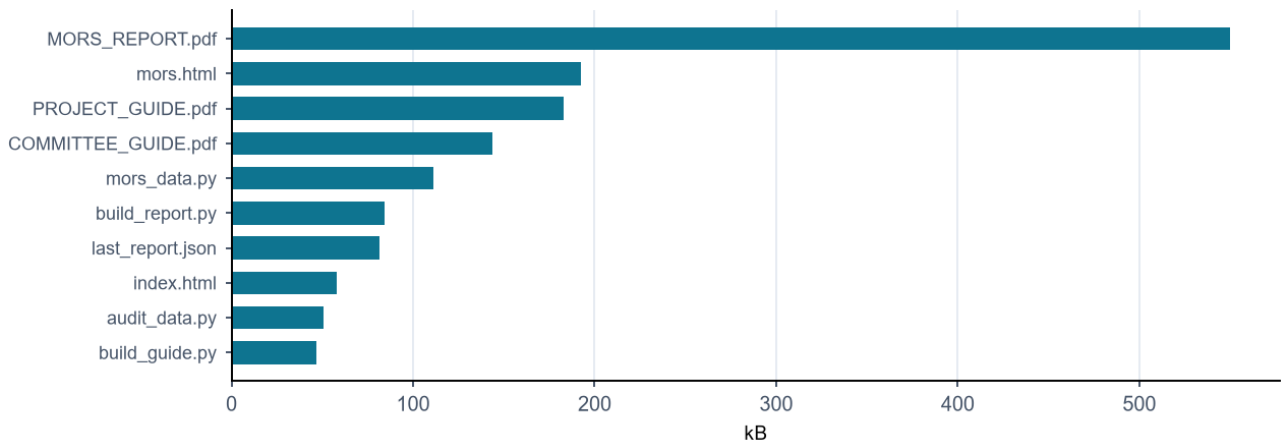


Figure 1 - Where the bytes are. The two HTML files and agents/mors_data.py dominate the tree; everything else is comparatively small. Large is not the same as complex - mors_data.py is one module per science view.

Folders

Folder	Files	kB	What lives there
project root	12	219.5	Entry points, secrets, packaging and the two human-readable manuals.
agents/	8	254.3	The four-agent pipeline plus the data layer behind MORS.
static/	2	250.5	The two front ends - no bundler, no framework, no build step.
tools/	5	228.2	The read-only data audit plus the document generators that turn the live API into PDFs.
prompts/	2	9.7	System prompts handed to the Gemini models.
data/	3	1.7	Small reference tables that keep the models honest offline.
docs/	4	898.0	Contract, report and this guide.

2. How a request flows end to end

Two front ends, one API, four agents.

Step	File	What happens
1	static/index.html or static/mors.html	The browser asks for a page and then for JSON.
2	main.py	GET /mors (or /) serves the HTML; GET /api/mors/<name> looks the module up in the _MORS dispatch table.
3	agents/mors_data.py (or sources_data.py)	The module runs its acquisition policy: live API -> Gemini -> local model, and stamps a trace badge on the payload.
4	agents/agent1_ingestion.py	On POST /api/pipeline/run: fetch, clean, audit, score.
5	agents/agent2_council.py	Five experts score the result and issue a verdict.
6	agents/agent3_citations.py	TRC ids, citations and provenance are attached.
7	agents/agent4_aimors.py	POST /api/chat turns the certified report into an answer that opens with the mandatory Arabic greeting.
8	docs/MORS_REPORT.pdf	tools/build_report.py reads the same API and freezes it into the evidence document.

Reading order for a reviewer

- README.md - how to start it in one command.
- Problems solved - the data/analysis defects found and fixed, with before/after numbers. The file carries an Arabic name (shown on the line below):

المشكلات_التي_تم_حلها.md

- docs/API_CONTRACT.md - what every endpoint returns.
- docs/MORS_REPORT.pdf - the evidence: quality, problems, figures, sources and limitations.
- docs/PROJECT_GUIDE.pdf (this file) - where each behaviour lives.
- DEPLOYMENT.md - how to run it somewhere else.
- tools/audit_data.py - 224 read-only checks -> RESULT: AUDIT: PASS.

3. Project root

Entry points, secrets, packaging and the two human-readable manuals.

00_COMMITTEE_GUIDE.md

Role	Kind	Size	Lines
Submission guide for the judging committee (Markdown)	Markdown document	27.1 kB	345

What it does

- Arabic walk-through of the package: what was built, how to score it in five minutes, the eight hackathon conditions, the sixteen scientific sources, the mandatory conventions and the self-test evidence.
- Same content as docs/COMMITTEE_GUIDE.pdf, in plain text so it can be read without a PDF reader.

Sections

دليل لجنة التحكيمية — ASI-HACK Space Analytics Engine

الفهرس

1. نظرة عامة على الحزمة

2. كيف تقيم المشروع خلال خمس دقائق

3. الملفات الرئيسية — كل ملف وما يفعله

4. المعمارية: خط أنابيب الوكيل الأربعة

5. أرقام حية يمكن للجنة التحقق منها

6. الشروط الثمانية — كيف تخدم المصادر الفريق

7. المصادر العلمية الستة عشر

8. الاتفاقيات الإلزامية الخمسة

9. التحقق الذاتي — أثبتتها بنفسك

=====

verdict : CERTIFIED

score : 98/100

Inputs and outputs

written by tools/build_committee.py.

Dependencies

reads the live API; falls back to cached numbers.

Where you see it

the first file listed in the package.

00_MANIFEST.txt

Role	Kind	Size	Lines
Package manifest (size + SHA-256 per file)	Text / prompt	13.1 kB	103

What it does

- Every file in the zip with its byte size, its SHA-256 digest and the role it plays, so a reviewer can prove the package was not altered in transit.

Inputs and outputs

written by tools/build_zip.py while the archive is packed.

Dependencies

hashes the files as they are compressed.

Where you see it

the second file listed in the package.

main.py

Role	Kind	Size	Lines
Flask orchestrator, HTTP API and CLI entry point	Python module	34.1 kB	813

What it does

- Builds the Flask application: loads .env, constructs the four agents, runs the pipeline when asked and serves every endpoint the two front ends call.
- Legacy ASI-HACK surface: GET /, GET /api/report, GET /api/data/<name> (space-weather, neo, horizons, neo-matrix, light-pollution, kp-index, spectral, quantum), POST /api/pipeline/run, GET /api/pipeline/status, GET /api/health.
- MORS surface: GET /mors serves the SPA, GET /api/mors lists the module registry, GET /api/mors/<name> dispatches into agents/mors_data.py, POST /api/mors/insight answers a grounded question, POST /api/chat runs Agent 4 (Ai.Mors).
- CLI: --selftest runs the whole pipeline offline and prints SELFTEST : PASS; --host/--port override the bind address.

Top-level symbols

```
def _mors_ai (103)
def _utc (124)
class State (128)
def _stage (184)
def run_pipeline (188)
def _pipeline_worker (268)
def _donki_series (278)
def _neo_series (304)
def _horizons_series (322)
def _neo_matrix (347)
def _science (381)
def create_app (397)
def selftest (727)
def _warm_satellites (752)
... +2 more
```

Inputs and outputs

reads .env, prompts/*.txt, last_report.json; writes HTTP JSON responses and the in-memory STATE report.

Dependencies

imports all four agents, agents.datasets, agents.mors_data and agents.sources_data; started by START.bat; called by tools/build_report.py, tools/build_guide.py and both HTML front ends.

Where you see it

the address bar of both dashboards and every API call listed in docs/API_CONTRACT.md.

START.bat

Role	Kind	Size	Lines
Windows launcher with environment checks	Batch script	4.3 kB	120

What it does

- Checks that python is on PATH, that .env exists and that the required packages import; frees the port if a previous instance is still bound; then starts main.py and opens the browser.
- Written with ``for /f "tokens=5"` to read the PID portably and saved with CRLF line endings so cmd.exe parses it correctly.

Inputs and outputs

reads requirements.txt and .env; spawns a python process.

Dependencies

invoked by a double-click; not used by the selftest.

Where you see it

the project root - the fastest way to run option 1.

requirements.txt

Role	Kind	Size	Lines
Python dependency list	Text / prompt	1.0 kB	30

What it does

- Pins the runtime libraries (google-generativeai, astropy, scikit-learn, pandas, numpy, requests, flask, python-dotenv) and lists the optional PDF stack (reportlab, arabic-reshaper, python-bidi, matplotlib).

Inputs and outputs

read by pip; read by START.bat.

Dependencies

required by every Python file in the project.

Where you see it

README.md step 0 and START.bat preflight.

README.md

Role	Kind	Size	Lines
Front door: how to run, what it does, what it does not do	Markdown document	15.7 kB	347

What it does

- Requirements, three run options (START.bat, manual, --selftest), the UI feature list, the API table, security notes, the project layout, how to regenerate the report, and an honest limitations section.

Sections

MORS - Scientific Command Center · ASI-HACK Space Analytics Engine · 1. Run locally · Option A - static site (no backend, mirrors the live demo exactly) · Option B - full backend (live Gemini, live NASA, real pipeline) · from the project folder · Option C - verify without a server · Option D - audit the data & analysis layer · Useful flags & configuration · Requirements · 2. Architecture & mission · 3. What the UI gives you · 4. Integrations (honest table) · 5. Tech stack

Inputs and outputs

read by humans and by anyone evaluating the submission.

Dependencies

documents main.py, both HTML files and the docs folder.

Where you see it

the first file a reviewer opens.

LICENSE

Role	Kind	Size	Lines
MIT license for the source code + data terms note	Configuration	2.0 kB	38

What it does

- Grants permission to use, copy, modify and distribute the code, and lists the third-party data terms (NASA, NOAA, CelesTrak) that still apply.

Inputs and outputs

read only.

Dependencies

referenced by README.md section 11.

Where you see it

root of the repository and of the delivered package.

المشكلات التي تم حلها.md

Role	Kind	Size	Lines
Problems solved - data & analysis defect log (Arabic)	Markdown document	23.1 kB	231

What it does

- Every defect found while building and auditing the engine, each with symptom, root cause (file:line), fix, before/after numbers and the audit check that now guards it.
- Includes the velocity unit bug, apogee/perigee zeros, the CelesTrak 403 fallback, CME/FLR counting, the habitability gate, the mixed Earth+Mars Horizons track, the cm-1 and Rabi unit errors, and the phase/period honesty fields.
- Ends with the four-member team roster and the commands needed to re-run the whole verification chain.

Sections

المشكلات التي تم حلها

(1) كيف تقرأ هذه الملفات

(2) مشكلات هذه الجولة (التركيز: البيانات والتحليل)

P-01 — (km/s معروضة كـ m/s) الوحدة الخاطئة لسرعة القمر الصناعي

P-02 — TLE عند غياب القيم في مصدر $\text{apogee/perigee} = 0$

P-03 — ومصدر الأقمار كله ميت (HTTP 403) يرفض الطلبات CelesTrak

P-04 — (FLR) كان يضم بقع الشمس (CME) عداد الانبعاثات الحلقية

P-05 — كان يحمل متوسط سرعة الانبعاثات الحلقية ``solar_wind_avg_kms``

P-06 — (FLR) إسقاط صامت لبعض الانبعاثات الشمسية

P-07 — كان صفراً لكل الكواكب `habitability_index` مؤشر صحة الأوج

P-08 — «Earth» مسار الأرض والمريخ مختلط داخل سلسلة واحدة باسم

P-09 — أعداد المدرجات الحرارية في الكمية خاطئة $\times 100$

P-10 — كان خطأ أفقياً عند الصفر (Rabi) منحني تنذب رابي

P-11 — ``mod 1`` تخالف تعريف `phase = 1.0`` مرحلة الطيف الضوئي

Inputs and outputs

written by hand; referenced by README.md and DEPLOYMENT.md.

Dependencies

quotes tools/audit_data.py check names.

Where you see it

project root, next to README.md.

DEPLOYMENT.md

Role	Kind	Size	Lines
Operations and deliverables manual	Markdown document	17.3 kB	356

What it does

- Environment variables, ports, the pipeline lifecycle, troubleshooting, the MORS UI section map, the theme engine, mandatory problem/Al conventions, how to regenerate both PDFs, and the deliverables checklist.

Sections

ASI-HACK Space Analytics Engine - Deployment & Operations Guide · 0. One-click launch (recommended) · 1. Manual launch · from the project folder · 2. Credentials (`.env`) · 3. Architecture · 4. API surface · 5. Security notes · 6. Rate limits & troubleshooting · 7. Validation · 7.1 Data & analysis audit · 8. Deploying beyond localhost (options) · 9. Hackathon deliverables map · 10. MORS Scientific Command Center (primary UI)

Inputs and outputs

read by an operator bringing the service up on another host.

Dependencies

references main.py routes and the docs folder.

Where you see it

the second document a reviewer opens.

.env.example

Role	Kind	Size	Lines
Template for the secrets file - no real keys	.example	0.3 kB	-

What it does

- Names the variables the backend reads (GEMINI_API_KEY, NASA_API_KEY, model ids) with placeholder values. Copy it to .env and fill in your own keys.

Inputs and outputs

shipped in the package; the real .env is gitignored and never packaged.

Dependencies

the template every deployment starts from.

Where you see it

never rendered anywhere - the UI only shows the engine name, never the key.

.gitignore

Role	Kind	Size	Lines
Keeps secrets and runtime artifacts out of the repository	(none)	0.3 kB	-

What it does

- Ignores .env, last_report.json, __pycache__/, *.pyc, *.log, virtualenvs and node_modules.

Inputs and outputs

read by git.

Dependencies

protects .env and last_report.json.

Where you see it

invisible at runtime.

last_report.json

Role	Kind	Size	Lines
Persisted copy of the most recent pipeline report	JSON document	81.3 kB	-

What it does

- Written after a successful pipeline run and read back when the process restarts, so /api/chat and tools/build_report.py still have a certified report to quote before the pipeline is re-run.

Inputs and outputs

written by main.py; read by main.py and tools/build_report.py.

Dependencies

produced by Agent 2 + Agent 3 output.

Where you see it

the health bar and the report PDF both fall back to it.

4. agents

The four-agent pipeline plus the data layer behind MORS.

agents/__init__.py

Role	Kind	Size	Lines
Package façade for the four agents	Python module	0.3 kB	10

What it does

- Re-exports IngestionAgent, CouncilAgent, CitationAgent and AiMorsAgent so main.py can import all of them from one place.

Inputs and outputs

none.

Dependencies

imported by main.py.

Where you see it

invisible - it is only a convenience module.

agents/agent1_ingestion.py

Role	Kind	Size	Lines
Agent 1 - Ingest & Clean	Python module	37.8 kB	816

What it does

- Fetches NASA DONKI (CME + flares), NASA NEOWS (near-Earth objects), JPL Horizons (heliocentric vectors), NASA GIBS night-light tiles and the NOAA SWPC planetary K-index, plus the three local reference CSVs.
- Cleans and types every column, flags duplicates, and runs the cell-level quality audit: completeness, accuracy, outlier count (StandardScaler + IsolationForest, contamination=0.1) and a single 0-100 data-quality score.
- Applies astropy Time/SkyCoord transforms to the Horizons vectors and a 3-sample moving average to the Kp series.

Top-level symbols

```
def _trace (48)
def _redact (61)
def _safe_get (74)
class IngestionAgent (84)
def _to_float (809)
```

Inputs and outputs

reads .env (NASA key) and data/*.csv; writes a rows/sources record into the in-memory report.

Dependencies

calls agents.datasets; called by main.py run_pipeline(); its quality dict is what DATA.MORS and the health bar show.

Where you see it

the four science widgets on the legacy dashboard, DATA.MORS, and section 3 of the report.

agents/agent2_council.py

Role	Kind	Size	Lines
Agent 2 - Council of Five	Python module	15.3 kB	330

What it does

- Runs a five-expert verification pass over the ingested dataset (data, science, engineering, QA, reproducibility) and turns their scores into one verdict with an overall score.
- Calls Gemini when quota allows and otherwise falls back to the deterministic local council; whichever engine answered is recorded as council.engine and printed in the header pill.

Top-level symbols

def _now (40)
def _quota_delay (44)
class CouncilAgent (56)

Inputs and outputs

reads prompts/council_prompt.txt and the Agent 1 record; writes council + metrics into the report.

Dependencies

depends on Agent 1; read by Agent 3, Agent 4, both UIs and the report cover.

Where you see it

the verdict badge on both dashboards and the cover page of MORS_REPORT.pdf.

agents/agent3_citations.py

Role	Kind	Size	Lines
Agent 3 - Citations & traceability	Python module	15.8 kB	313

What it does

- Builds the TRC-### traceability ids, the citation list and the provenance graph that links every figure back to a NASA/NOAA endpoint and a processing step.
- Runs Gemini citation enrichment when quota allows and degrades to the deterministic baseline otherwise.

Top-level symbols

class CitationAgent (211)

Inputs and outputs

reads the Agent 1 + Agent 2 records; writes citations into the report.

Dependencies

depends on Agents 1 and 2; consumed by Agent 4 (it quotes the trace ids) and by section 11 of the report.

Where you see it

the [TRACE] badges everywhere and the citations panel on AI.MORS.

agents/agent4_aimors.py

Role	Kind	Size	Lines
Agent 4 - Ai.Mors conversational agent	Python module	12.7 kB	251

What it does

- Answers free-text questions about the certified report. It has a hard constraint: the reply must open with - if the model omits it the agent prepends it.
- Engine ladder: gemini:3.1-pro -> gemini:3.6-flash -> gemini-flash-latest -> deterministic offline template. The rung that answered is echoed back in the `engine` field.
- The offline template answers in numbered Arabic steps with the trace ids, so the UI is never empty even with no quota.

Top-level symbols

class AiMorsAgent (28)

Inputs and outputs

reads prompts/aimors_prompt.txt, the report and the chat history; returns reply/greeting_ok/trace/engine/timestamp.

Dependencies

instantiated per request by POST /api/chat in main.py; used by the legacy chat card and the AI.MORS chat box.

Where you see it

the Ai.Mors chat card on index.html and the 'Ask Ai.Mors directly' box on AI.MORS.

agents/datasets.py

Role	Kind	Size	Lines
Dataset builders for the science widgets	Python module	32.4 kB	754

What it does

- Produces the light-pollution / sky-brightness model from NASA GIBS VIIRS and Black-Marble tiles, the spectral reference matrix, the reference metrics table and the Kp + flare series.
- Every builder records where the numbers came from and whether they are observed, calculated or modelled.

Top-level symbols

```
def _cached (51)
def _store (59)
def _utc (65)
def _bortle_from_nelm (119)
def _fetch_tile (129)
def fetch_light_pollution (154)
def light_pollution_reference_csv (262)
def fetch_kp_index (299)
def _fetch_flares (415)
def _fallback_flares (469)
def _fallback_kp (485)
def build_spectral (520)
def _write_spectral_csv (603)
def run_quantum (622)
... +3 more
```

Inputs and outputs

reads data/*.csv; calls NASA GIBS and NOAA SWPC.

Dependencies

used by Agent 1 and by the light-pollution / spectral endpoints in main.py.

Where you see it

the legacy dashboard charts and PROJECT PRJ-002.

agents/mors_data.py

Role	Kind	Size	Lines
MORS data layer - 19 API modules	Python module	111.2 kB	2 260

What it does

- One module per MORS view: data_health, problems, solutions, home, photometry, spectroscopy, lightcurves, images, exoplanets, objects, fits_headers, satellites, quantum_lab, ai_workspace, projects, team, modules_catalog and ai_insight.
- Implements the acquisition policy every module obeys: live API first, Gemini synthesis second, deterministic local model last, with the chosen rung recorded in trace.via.
- Owns the trace()/badge() helpers that stamp [Source | Version | Pipeline | Method | Updated | Operator] on every payload, and a 300 s TTL cache.

Top-level symbols

def configure (54)
def _iso (62)
def trace (66)
def badge (85)
def _cached (93)
def _put (101)
def _key_lock (109)
def _get_json (117)
def _ai_json (127)
def _snapshot_cache (153)
def acquire (190)
def reset (241)
def data_health (267)
def _p (309)
... +40 more

Inputs and outputs

calls CelesTrak, NASA APOD, the Exoplanet Archive TAP, DONKI, NEOWS, Horizons, GIBS and NOAA; reads the report.

Dependencies

dispatched by GET /api/mors/<name>; imported by tools/build_report.py and agents/sources_data.py.

Where you see it

every view of the MORS SPA except the legacy dashboard.

agents/sources_data.py

Role	Kind	Size	Lines
Scientific source registry and hackathon condition map	Python module	28.8 kB	528

What it does

- Holds the 16 official ASI Hackathon sources (phase, level, Arabic guidance, link, and the file in this repo that uses them) and the 12 live data sources this build actually calls.
- Holds the eight 'how do the sources help the team?' conditions, each with its Arabic wording, its supporting source ids and its evidence inside the repository.
- build() returns the payload served at GET /api/mors/sources.

Top-level symbols

def build (465)

Inputs and outputs

static - no network access, so it renders fully offline.

Dependencies

imported by main.py; feeds SOURCES.MORS and section 15 of MORS_REPORT.pdf.

Where you see it

the SOURCES.MORS view and section 15 of the report.

5. static

The two front ends - no bundler, no framework, no build step.

static/index.html

Role	Kind	Size	Lines
Legacy ASI-HACK dashboard (single file, no build step)	Single-file web app	58.0 kB	1 207

What it does

- The original submission UI: pipeline run button, four-agent stage log, science widgets for space weather, near-Earth objects, light pollution and spectra, a Three.js star field, and the Ai.Mors chat card.
- Self-contained: CSS, JavaScript and the import map for three.js all live inside the one file, so it works from a plain static server.

Title	script blocks	style blocks	element ids
ASI-HACK . Space Analytics Engine - Ai.Mors	5	1	38

Inputs and outputs

fetches /api/report, /api/data/*, POST /api/pipeline/run and POST /api/chat.

Dependencies

served by GET / in main.py; shares the same API as the MORS SPA.

Where you see it

<http://127.0.0.1:5000/>

static/mors.html

Role	Kind	Size	Lines
MORS Scientific Command Center SPA	Single-file web app	192.5 kB	3 332

What it does

- A single-page application with a sidebar router: MORS Home, AI.MORS, DATA.MORS, ASTRONOMY / SATELLITE / QUANTUM, Problems -> Solutions, Solutions Center, PROJECTS, SOURCES and TEAM.
- Chart.js visualisations, a global search index over every payload, theme engine (dark/light + six planet accents) stored in localStorage under mors.settings.v1, and a health bar that re-reads data health on every view.
- The SOURCES view renders the 16 official sources with their Arabic guidance and the eight-condition compliance matrix; the AI view explains who Ai.Mors is and offers a live chat box.

Title	script blocks	style blocks	element ids
MORS . Scientific Command Center	3	1	66

Inputs and outputs

fetches /api/mors/<name> for every view plus /api/chat and /api/mors/insight.

Dependencies

served by GET /mors in main.py; Chart.js comes from cdn.jsdelivr.net (allowed by the CSP).

Where you see it

<http://127.0.0.1:5000/mors>

6. tools

The read-only data audit plus the document generators that turn the live API into PDFs.

tools/audit_data.py

Role	Kind	Size	Lines
Read-only data & analysis audit (224 checks)	Python module	50.9 kB	1 090

What it does

- Probes the running API and prints one line per check; exits 0 only when every check passes (RESULT: AUDIT: PASS).
- Covers the priority formula and its 75/50 labels, Kepler period /velocity agreement, apogee/perigee integrity, CME-vs-FLR counting, habitability-gate reproduction, per-body Horizons radii, quantum normalisation and unit conversions, spectral PCA, light-pollution calibration direction and the team roster.
- Batch 2 added physics checks (horizons, exoplanet gates, quantum, spectral, light pollution, curves, Kp daily, images).

Top-level symbols

```
def get (39)
def check (47)
def near (55)
def mean (62)
def same_utc_day (67)
def counts_agree (80)
def load (93)
def audit_health (133)
def audit_datahealth (144)
def audit_problems (179)
def audit_solutions (241)
def audit_satellite (260)
def audit_exoplanets (336)
def audit_objects (356)
... +18 more
```

Inputs and outputs

reads HTTP only; writes nothing.

Dependencies

needs a running server (python main.py); used before every document regeneration.

Where you see it

RESULT: AUDIT: PASS on the console.

tools/build_report.py

Role	Kind	Size	Lines
Generates docs/MORS_REPORT.pdf	Python module	84.2 kB	1 901

What it does

- Fetches every module from the running server (or falls back to last_report.json), sanitises text for reportlab, and lays out a 16-section dossier: cover, contents, executive summary, architecture, data-health audit, problem cards, solutions, module registry, dataset inventory, research dossiers, team, AI protocol, traceability, API reference, run instructions, figures, scientific sources and honest limitations.
- Arabic support: arabic_reshaper + python-bidi + Arial, wrapped word by word so reportlab never re-wraps a pre-reversed line.
- Chart helpers (matplotlib, Agg) render the six figures used in section 14 and 15 as PNGs embedded in the PDF.

Top-level symbols

```
def _clean (101)
def _xml (121)
def san (125)
def own (133)
def san_ml (142)
def _now (158)
def _num (162)
def n_of (169)
def _p (187)
def P (242)
def R (246)
def table (251)
def kpis (279)
def bullets (305)
... +24 more
```

Inputs and outputs

reads the live API or last_report.json; writes docs/MORS_REPORT.pdf.

Dependencies

needs reportlab (+ optional arabic-reshaper, python-bidi, matplotlib); imported by tools/build_guide.py for its shared styles.

Where you see it

docs/MORS_REPORT.pdf

tools/build_guide.py

Role	Kind	Size	Lines
Generates this document (docs/PROJECT_GUIDE.pdf)	Python module	47.0 kB	986

What it does

- Walks the repository, parses each Python file with the ast module to list its top-level symbols, and prints a card for every file: role, what it does, inputs/outputs, dependencies and where the reader sees its effect.
- Shares styles, tables and the file-size chart with tools/build_report.py by importing it.

Top-level symbols

```
def _read (636)
def _py_symbols (645)
def _md_headings (659)
def _html_facts (669)
def _csv_facts (684)
def scan (691)
def _kind (722)
def _symbol_line (731)
def _has_arabic (741)
def _path_para (745)
def card (755)
def build (800)
def main (971)
```

Inputs and outputs

reads the working tree; writes docs/PROJECT_GUIDE.pdf.

Dependencies

requires reportlab, arabic-reshaper, python-bidi and matplotlib.

Where you see it

docs/PROJECT_GUIDE.pdf

tools/build_committee.py

Role	Kind	Size	Lines
Generates 00_COMMITTEE_GUIDE.md and docs/COMMITTEE_GUIDE.pdf	Python module	41.3 kB	814

What it does

- One content model rendered twice - Markdown for quick reading, PDF for printing - so the two can never drift apart.

Top-level symbols

```
def is_ar (49)
def api (57)
def collect (66)
def n (87)
def kb (95)
def _pad (99)
def cell (103)
def header_cell (111)
def make_table (119)
def para (144)
def code_block (149)
def p (170)
def bul (171)
def tab (172)
... +14 more
```

Inputs and outputs

reads the live API and the file registry; writes the two committee documents.

Dependencies

imports tools/build_report.py for styles and the Arabic text helpers, and tools/build_guide.py for the file registry.

Where you see it

regenerate with: python tools/build_committee.py

tools/build_zip.py

Role	Kind	Size	Lines
Portable packager - writes the submitted archive	Python module	4.7 kB	147

What it does

- Zips exactly the files the judges receive: sources, docs, datasets, prompts, the Arabic problems log and the generator scripts, skipping __pycache__, .git and virtualenvs.
- Resolves the repository root from __file__, so it runs from any working directory; prints entry count, raw size and zipped size.

Top-level symbols

```
def digest (96)
```

Inputs and outputs

reads the project tree; writes ASI-HACK-Space-Analytics-Engine.zip at the project root.

Dependencies

stdlib zipfile only; run last, after audit_data.py and the three document builders.

Where you see it

ASI-HACK-Space-Analytics-Engine.zip and 00_MANIFEST.txt.

7. prompts

System prompts handed to the Gemini models.

prompts/council_prompt.txt

Role	Kind	Size	Lines
System prompt for Agent 2 (Council of Five)	Text / prompt	4.9 kB	98

What it does

- Instructs the five experts on how to score the dataset, what a defensible verdict looks like, and that fabricated sources are an automatic failure.

Inputs and outputs

read at each council call; never modified at runtime.

Dependencies

used by agents/agent2_council.py.

Where you see it

its effect is the verdict badge on the dashboard.

prompts/aimors_prompt.txt

Role	Kind	Size	Lines
System prompt for Agent 4 (Ai.Mors)	Text / prompt	4.8 kB	87

What it does

- Defines the persona, the numbered-step answer format, the mandatory Arabic greeting, and the rule that answers may only be grounded in the certified report.

Inputs and outputs

read at each chat call.

Dependencies

used by agents/agent4_aimors.py.

Where you see it

the replies in both chat boxes.

8. data

Small reference tables that keep the models honest offline.

data/light_pollution_reference.csv

Role	Kind	Size	Lines
Reference night-light / sky-brightness table	CSV dataset	0.6 kB	10

What it does

- Eleven sites with radiance, sky brightness and Bortle-class reference values used to calibrate the light-pollution model.

Rows	Columns
9	6

Inputs and outputs

read by agents/datasets.py and Agent 1.

Dependencies

feeds PRJ-002 and the light-pollution problem card when its coverage condition is open.

Where you see it

the light-pollution widget and the DATA.MORS table.

data/reference_metrics.csv

Role	Kind	Size	Lines
Reference metrics for the quality audit	CSV dataset	0.6 kB	12

What it does

- Expected ranges and units used when the ingestion layer decides whether a cell is valid.

Rows	Columns
11	4

Inputs and outputs

read by Agent 1 during cleaning.

Dependencies

feeds the accuracy score.

Where you see it

the completeness / accuracy tiles.

data/spectral_reference.csv

Role	Kind	Size	Lines
Laboratory emission-line list	CSV dataset	0.4 kB	7

What it does

- Wavelengths and intensities used to build the synthetic spectra shown in the spectroscopy module.

Rows	Columns
6	9

Inputs and outputs

read by agents/datasets.py.

Dependencies

feeds the spectral widget and P-003.

Where you see it

ASTRONOMY.MORS spectroscopy and the legacy spectral chart.

9. docs

Contract, report and this guide.

docs/API_CONTRACT.md

Role	Kind	Size	Lines
Machine-readable-ish API contract	Markdown document	21.5 kB	462

What it does

- Sections 1-14 document the legacy endpoints with example request/response bodies; sections 15-19 document the MORS UI, every /api/mors module schema and the health bar.

Sections

ASI-HACK Space Analytics Engine — Shared API Contract (Backend ☐ Frontend)

كل نقطة نهاية إلزامية للتكامل بين static/index.html و main.py

1) GET / → يقدم static/index.html

2) GET /api/health

3) POST /api/pipeline/run (تشغيل المخطط الكامل: Agent1 → Agent2 → Agent3)

4) GET /api/pipeline/status

5) GET /api/report → آخر تقرير معتمد (يخزن في حالة الذاكرة + last_report.json)

6) GET /api/data/space-weather (DONKI — للرسوم البيانية)

7) GET /api/data/neo (NEOWS)

8) GET /api/data/horizons (JPL Horizons — إحداثيات جرم)

9) GET /api/data/neo-matrix (NEO hazard bubble matrix → widget 2)

10) GET /api/data/light-pollution (NASA GIBS VIIRS Black Marble → widget 1)

11) GET /api/data/kp-index (NOAA SWPC Kp + NASA DONKI flares → widget 3)

12) GET /api/data/spectral (Sentinel-2 band indices → widget 4)

Inputs and outputs

read by a client author.

Dependencies

mirrors main.py - keep both in sync.

Where you see it

the reference used when adding a new endpoint.

docs/MORS_REPORT.pdf

Role	Kind	Size	Lines
Generated platform report (44 pages)	Generated PDF	550.1 kB	-

What it does

- The evidence document: quality audit, problem and solution cards, module registry, figures with plain-English captions, the full scientific source list and the honest limitations.

Inputs and outputs

written by tools/build_report.py.

Dependencies

requires a running server (or last_report.json).

Where you see it

the submission package.

docs/PROJECT_GUIDE.pdf

Role	Kind	Size	Lines
This document	Generated PDF	182.7 kB	-

What it does

- A file-by-file explanation of the repository so a reviewer can find any behaviour in one lookup.

Inputs and outputs

written by tools/build_guide.py.

Dependencies

requires reportlab and the optional PDF stack.

Where you see it

the submission package.

docs/COMMITTEE_GUIDE.pdf

Role	Kind	Size	Lines
Submission guide for the judging committee (PDF)	Generated PDF	143.7 kB	-

What it does

- The printable Arabic edition of 00_COMMITTEE_GUIDE.md, with tables, KPI cards and page numbers.

Inputs and outputs

written by tools/build_committee.py.

Dependencies

requires reportlab, arabic-reshaper and python-bidi.

Where you see it

the submission package.

10. Regenerating the documents

Command	Output
python main.py	server on http://127.0.0.1:5000
python main.py --selftest	offline pipeline check, prints SELFTEST : PASS
python tools/build_report.py	docs/MORS_REPORT.pdf (needs the server running)
python tools/build_guide.py	docs/PROJECT_GUIDE.pdf (this file)

Both generators read the live API first and fall back to last_report.json, so they work either way. The optional PDF stack is listed at the bottom of requirements.txt.

Guide generated 2026-09-26T07:53:51+00:00 from C:\Users\sha\Desktop\ (Projects)\New folder (6) - 36 files scanned.